

Unit-2

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→ Propositional Logic :

Defn, Statements & ~~logic~~ Notation;
Truth values, well Formed Formulas,
Tautologies, Equivalence of Formulas,
Duality Law, Tautological implications,
Examples.

→ Predicate Logic :

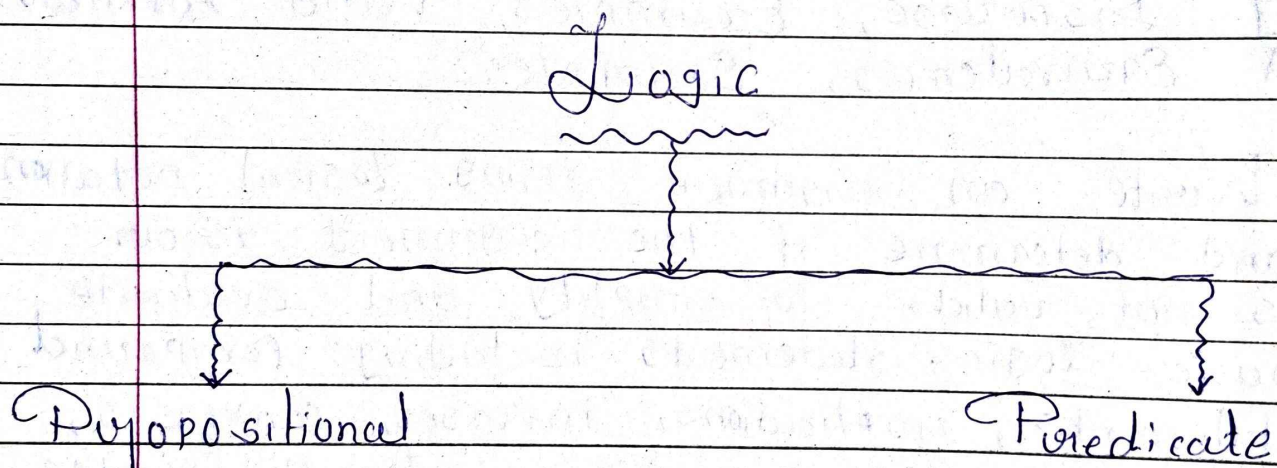
Defn. of Predicates, statement functions,
Variables, Quantifiers, Predicate Formulas,
Free & Bound Variables, The universe
of Discourse, Examples, Valid Formulas
& Equivalences, Examples.

Co-2 : Write an argument using logical notation and determine if the argument is or is not valid. To simplify and evaluate basic logic statements including compound statements, implications, inverses, converses, and Contra positive using truth Tables and the properties of Logic. To express a logic sentence in terms of predicates, quantifiers and logical connectives.

Propositional Logic

Logic is concerned with methods of reasoning. Aristotle (381-322 BC) first started studying logical reasoning.

Logic have provided the theoretical base for many areas of Computer science such as digital logic design, automata theory, & computability, artificial intelligence etc.



[A] Propositional Logic :-

Defn :- A number of words making a complete grammatical structure having a sense and meaning and also meant an assertion in logic or mathematics is called a Sentence.

Defn :- A declarative sentence which is either true or false is called Proposition or Statement.

ex : "Three plus three equals Six"

ex : "Three plus three equals Nine"

→ The truth or falsity of a stmt. is called truth value.

→ Simple stmts. are generally represented by letter $P, q, r, \dots$ etc. known as proposition variables.

★ Connectives

Symbol	Word	Nature	Symbolic form
$\sim, \neg, \bar{}$	not	Negation	$\sim P$
$\wedge$	and	Conjunction	$P \wedge Q$
$\vee$	Or	Disjunction	$P \vee Q$
$\Rightarrow, \rightarrow$	If ---- Then	Implication (conditional)	$P \Rightarrow Q$
$\Leftrightarrow$	If and only if	Equivalence (Bi-Conditional)	$P \Leftrightarrow Q$

Negation :-

If P is any proposition, the negation of P is denoted by $\sim P$. $\sim$ is a propo. which is False, when P is True & True when P is False.

Ex : P : All students are intelligent.

Then

$\sim P$: Some students are not intelligent or

$\sim P$: There exists a student who is not intelligent.

Truth Table :-

P	$\sim P$
T	F
F	T

Conjunction :-

If P & Q are two statements then Conjunction of P and Q is denoted by $P \wedge Q$ and read as 'P and Q'.

→ $P \wedge Q$ is True only when both P & Q are True, otherwise it is False.

Ex If P : Ram is healthy & Q : He has blue eyes.
Then form $P \wedge Q$.
 $P \wedge Q$: Ram is healthy & he has blue eyes.

Truth Table :-

P	q	$P \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction :-

If P & q are two statements then the disjunction of P and q is denoted by $P \vee q$ and read as 'P or q'.

→ $P \vee q$ is true only when at least one from P or q is True.

ex: P : It is cold ; q : It is raining
Form $P \vee q$.

$P \vee q$: It is cold or raining.

Truth Table :-

P	q	$P \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Conditional Proposition :-

If p & q are propositions, then "the compound proposition "if p then q " denoted by $p \Rightarrow q$ (or $p \rightarrow q$) is called a conditional propo. or implication.

Here p is called antecedent or hypothesis & q is called consequent or conclusion.

$p \Rightarrow q$ is False when p is True & q is False, otherwise it's - True.

Ex :- p : It rains; q : I will carry an umbrella

Then $p \Rightarrow q$: If it rains then I will carry an umbrella.

Truth Table :-

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

NOTE :- If $p \Rightarrow q$ is an implication then

- (i) $q \Rightarrow p$ is called its converse;
- (ii) $\sim q \Rightarrow \sim p$ is called its contrapositive;
- (iii) $\sim p \Rightarrow \sim q$ is called its inverse.

Biconditional Statement :-

If P and q are statement then the compound statement ' P if and only if q ' denoted by $P \Leftrightarrow q$ is called a biconditional statement.

The truth values of $P \Leftrightarrow q$ is True when either both P & q are True or both P & q are False.

Ex: P : A new car will be acquired
 q : Additional Funding is available.

Then $P \Leftrightarrow q$: A new car will be acquired if and only if additional Funding is available.

Truth Value :

P	q	$P \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Tautologies and Contradictions

" A Compound proposition that is always true, for all possible truth values of its variables is called a Tautology "

" A Compound proposition that is always False, for all possible truth values of its variables is called a Contradiction "

" A proposition that is neither a Tautology nor a contradiction is called contingency.

[Ex] People either like watching TKS or they don't. $\Leftarrow$ Tautology

[Ex] All men are good and all men are ~~men~~ bad $\Leftarrow$ Contradiction.

Logical Equivalence

Two propositions A and B are Logically equivalent if and only if they have the same truth values for every choice of

Problems based on Propositional Logic

Q-1: Construct a truth table for the following

- (1) $P \wedge (q \vee q)$
- (2) $\sim (P \vee q) \vee (\sim P \wedge \sim q)$
- (3) $P \vee \sim q \Rightarrow P$
- (4) $[\sim (P \wedge q) \vee q] \Rightarrow \sim P$

Q-2: Use Truth Table to prove the following equivalence.

- (1) $P \vee (q \wedge r) \equiv (P \vee q) \wedge (P \vee r)$
- (2) $P \Rightarrow q \equiv \sim P \vee q$
- (3) $\sim q \Rightarrow \sim P \equiv P \Rightarrow q$
- (4) $P \Leftrightarrow q \equiv (P \Rightarrow q) \wedge (q \Rightarrow P)$
- (5) $P \Leftrightarrow q \equiv (P \vee q) \Rightarrow (P \wedge q)$

Q-3: Prove that the following are tautology.

- (1) $P \vee \sim P$
- (2) $\sim (P \wedge q) \vee q$
- (3) $P \Rightarrow (P \vee q)$

Q-4: Verify that the proposition $P \wedge (q \wedge \sim P)$ is a Contradiction.

Q-5: Prove that the statement below is Contingency:
 $(P \rightarrow q) \rightarrow (P \wedge q)$

Q-6: Determine whether the following is a Tautology, Contingency or Contradiction

- (i) $P \rightarrow (P \rightarrow q)$ [Conti.]
- (ii) $P \rightarrow (q \rightarrow P)$ [Taut.]
- (iii) $P \wedge \sim P$ [Contrad.]

Duality Law :

Two formulae A and A^* are said to be the duals of each other if either one can be obtained from the other by replacing ' $\wedge$ ' by ' $\vee$ ' and ' $\vee$ ' by ' $\wedge$ '.

Ex-1

$$(i) \sim (p \vee q) \equiv \sim p \wedge \sim q \quad \&$$
$$(ii) \sim (p \wedge q) \equiv \sim p \vee \sim q$$

Dual of (i) is (ii) & vice versa

Predicate Logic

e.g. $\rightarrow x < 4, x+y \leq 5, x+y+z = 10$

As we don't know the values of x, y and z , above are sentences but not statements.

Such statements are called Predicates.

★ Universal Quantifier :-

If $P(x)$ is a predicate with the variable x as an argument, then the mathematical symbol for the quantifier "for all", denoted by ' $\forall$ ', can be written as " $\forall x, P(x)$ ".

★ Existential Quantifiers :-

Suppose for the predicate $P(x)$, $\forall x, P(x)$ is false, but there exists at least one value of x for which, $P(x)$ is true, then we say that, x is bounded by existential quantification. It is denoted by $\exists$.

Hence $\forall x [P(x)] \equiv \exists x [-P(x)]$

Ques 1: Rewrite using quantifiers & predicate symbols

(1) All birds can fly.

$P(x)$: x is a bird

$Q(x)$: x can fly.

$$\forall x [P(x) \rightarrow Q(x)]$$

(2) Some men are genius.

$A(x)$: x is a man;

$B(x)$: x is a genius.

$$\exists x [A(x) \wedge B(x)]$$

(3) Some numbers are not irrational.

$P(x)$: x is a number

$Q(x)$: x is irrational

$$\exists x [P(x) \wedge \sim Q(x)]$$